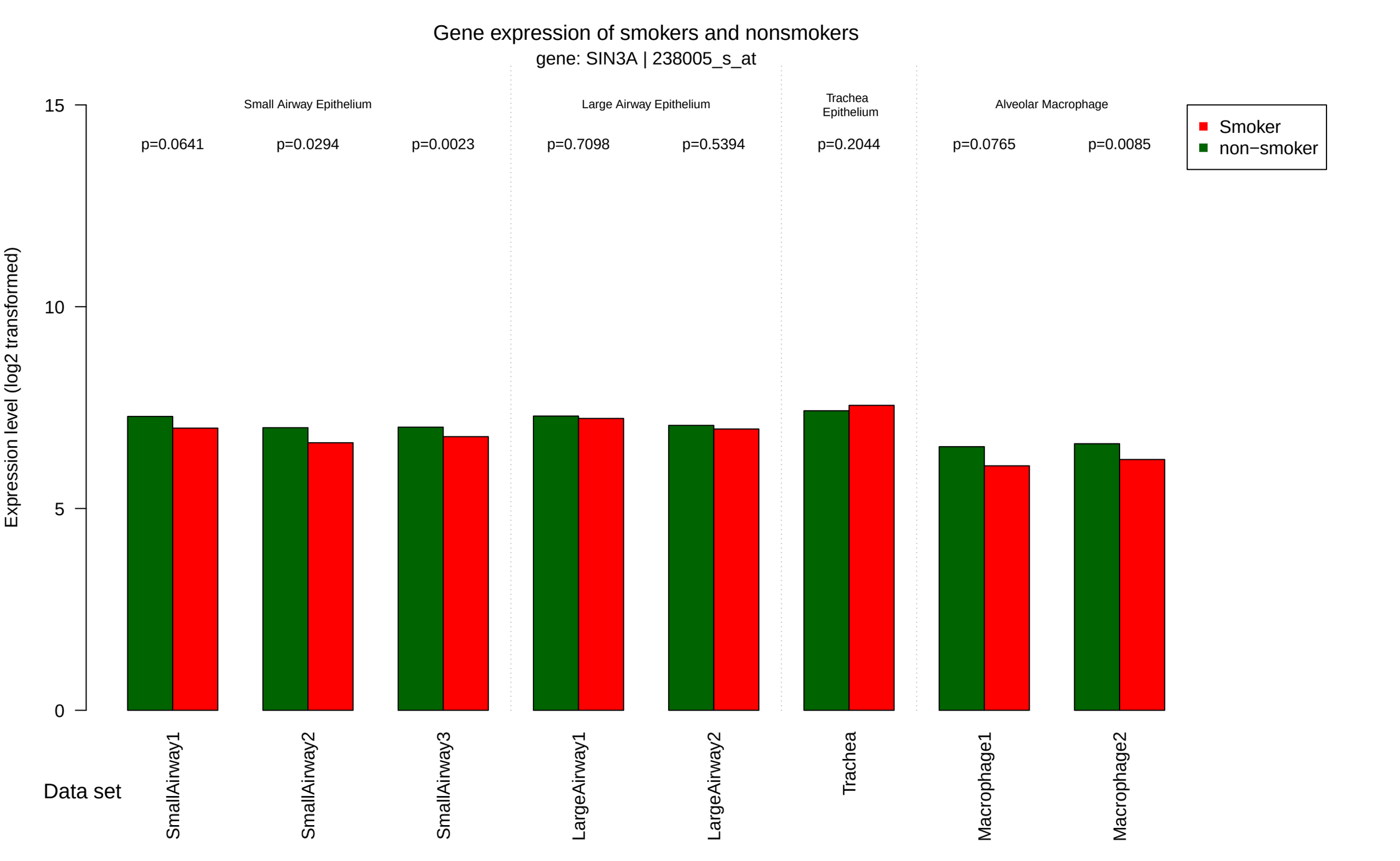
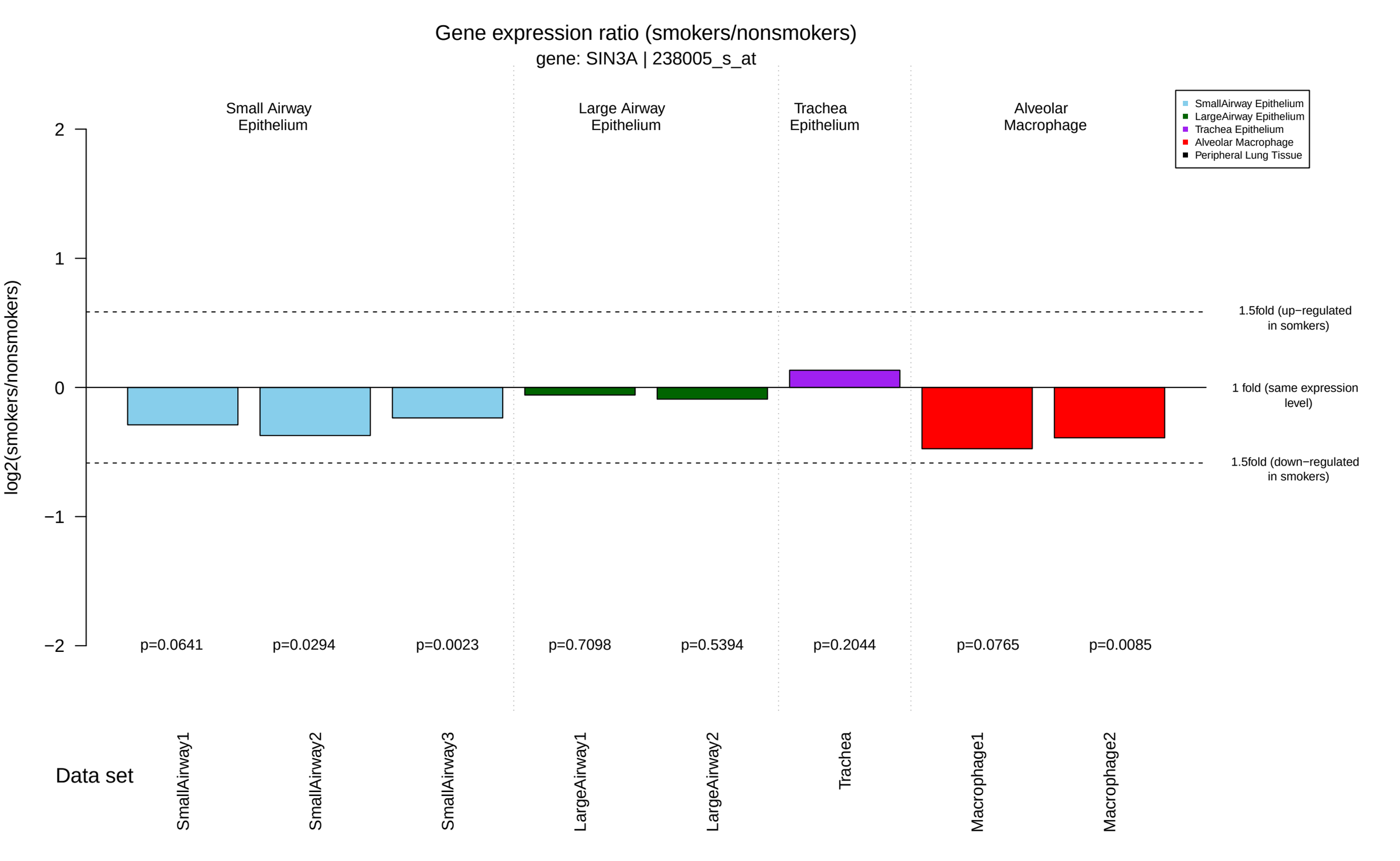
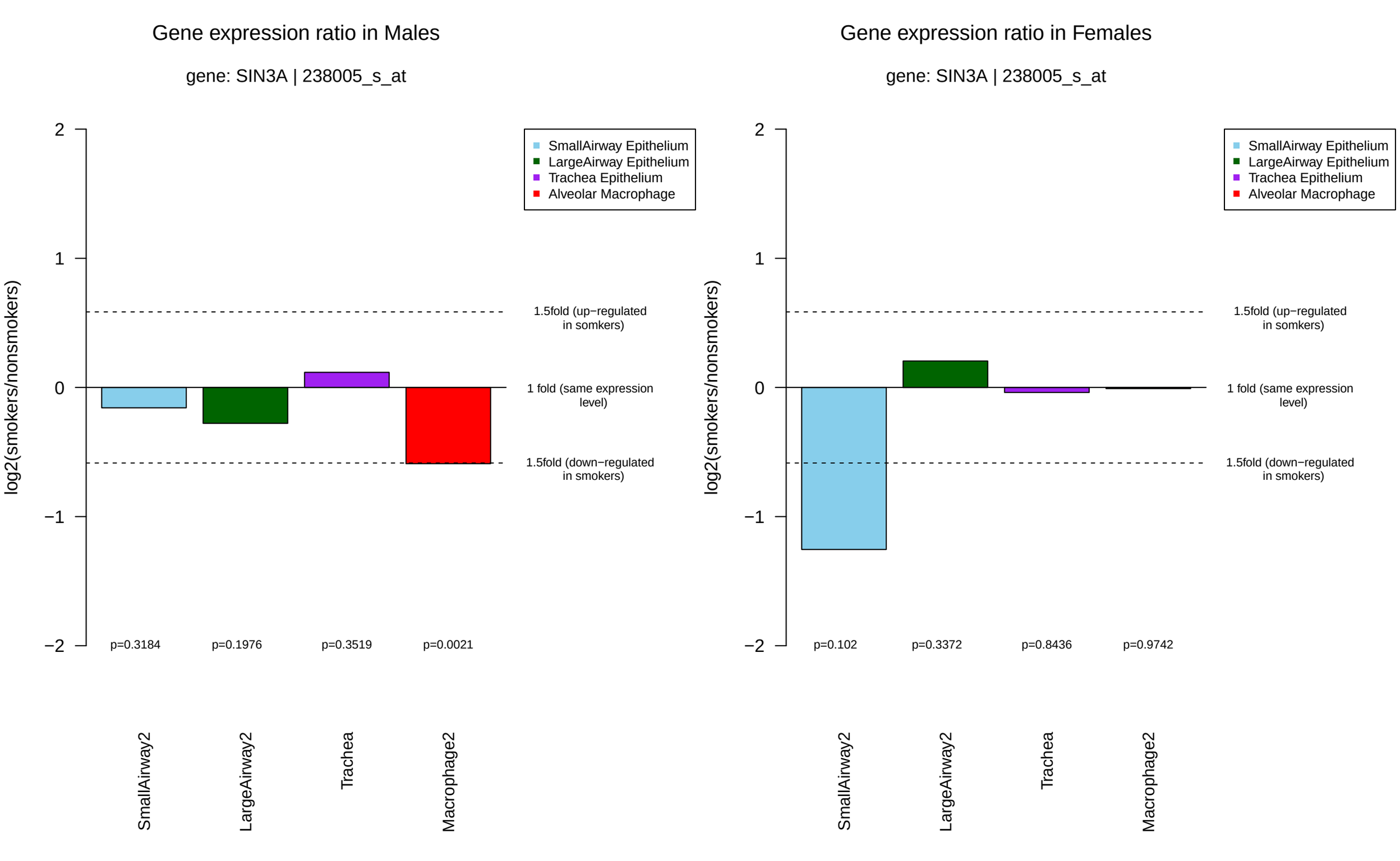
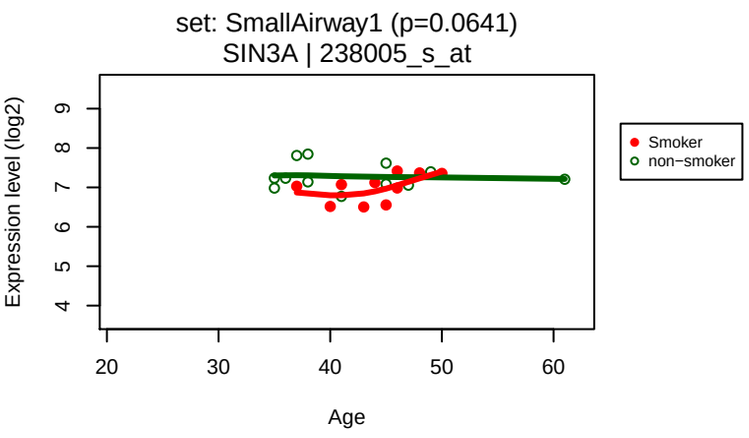
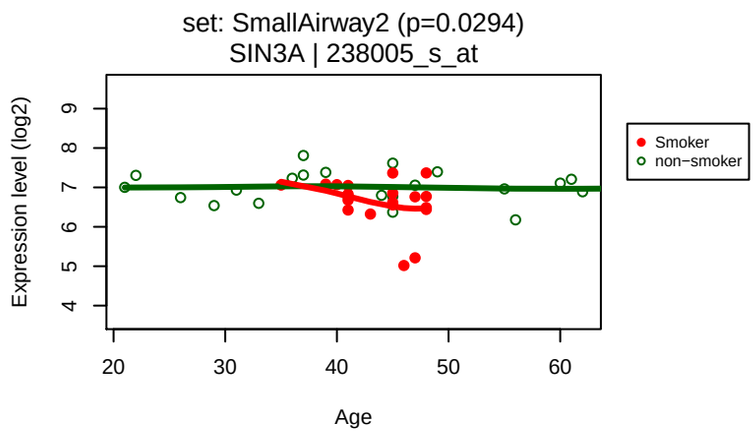
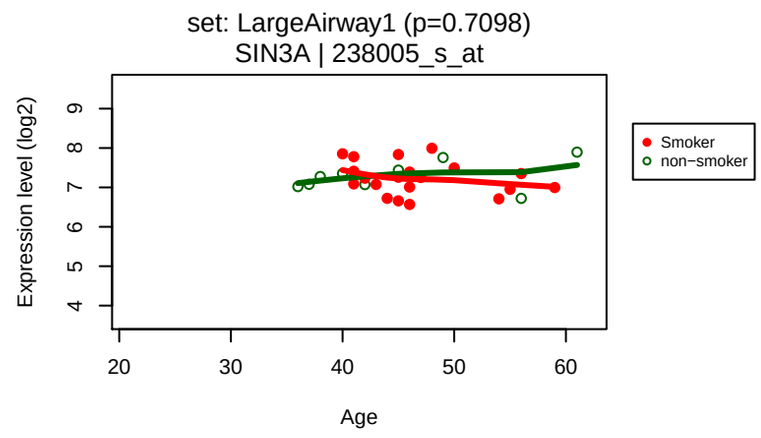

Gene expression of smokers and nonsmokers
gene: SIN3A | 238005_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0641
p=0.0294
p=0.0023
p=0.7098
p=0.5394
p=0.2044
p=0.0765
p=0.0085
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

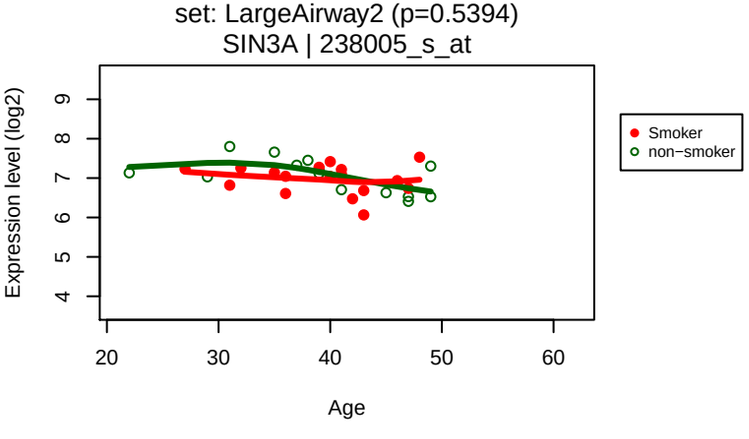

Gene expression ratio (smokers/nonsmokers)
gene: SIN3A | 238005_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0641
p=0.0294
p=0.0023
p=0.7098
p=0.5394
p=0.2044
p=0.0765
p=0.0085
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

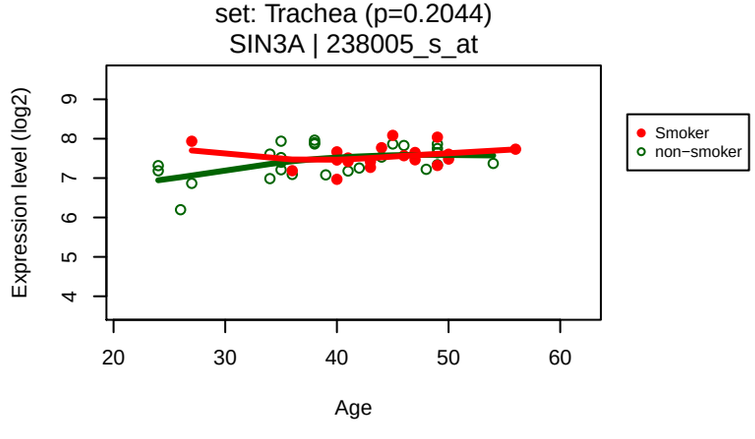

Gene expression ratio in Males
gene: SIN3A | 238005_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3184
p=0.1976
p=0.3519
p=0.0021
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SIN3A | 238005_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.102
p=0.3372
p=0.8436
p=0.9742
SmallAirway2
LargeAirway2
Trachea
Macrophage2

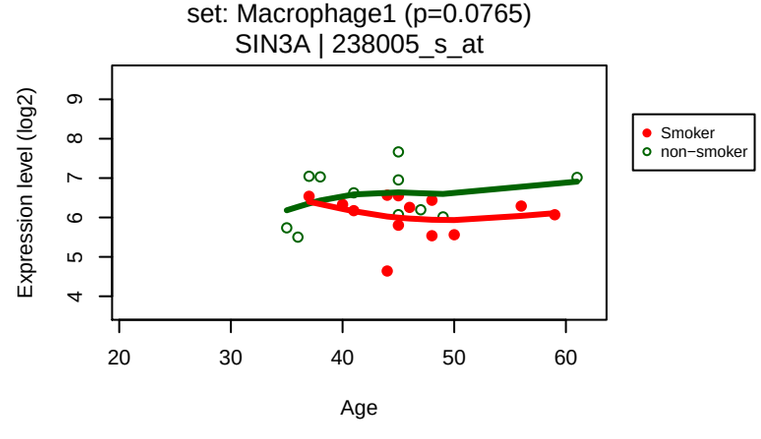

set: SmallAirway1 (p=0.0641)
SIN3A | 238005_s_at
Expression level (log2)
Age
Smoker
non−smoker

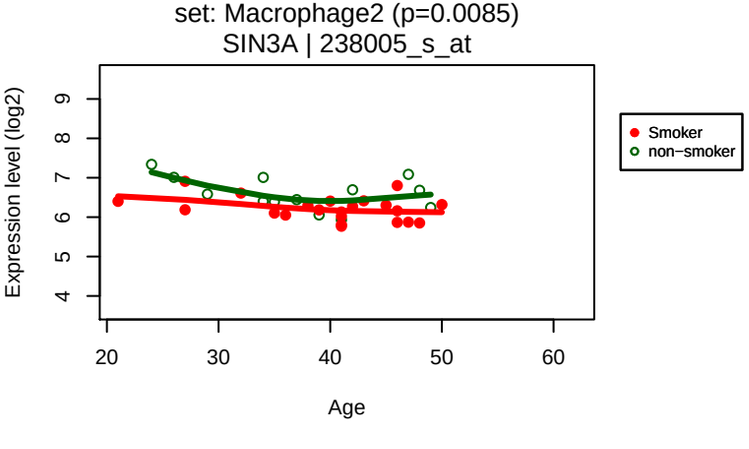

set: SmallAirway2 (p=0.0294)
SIN3A | 238005_s_at
Expression level (log2)
Age
Smoker
non−smoker

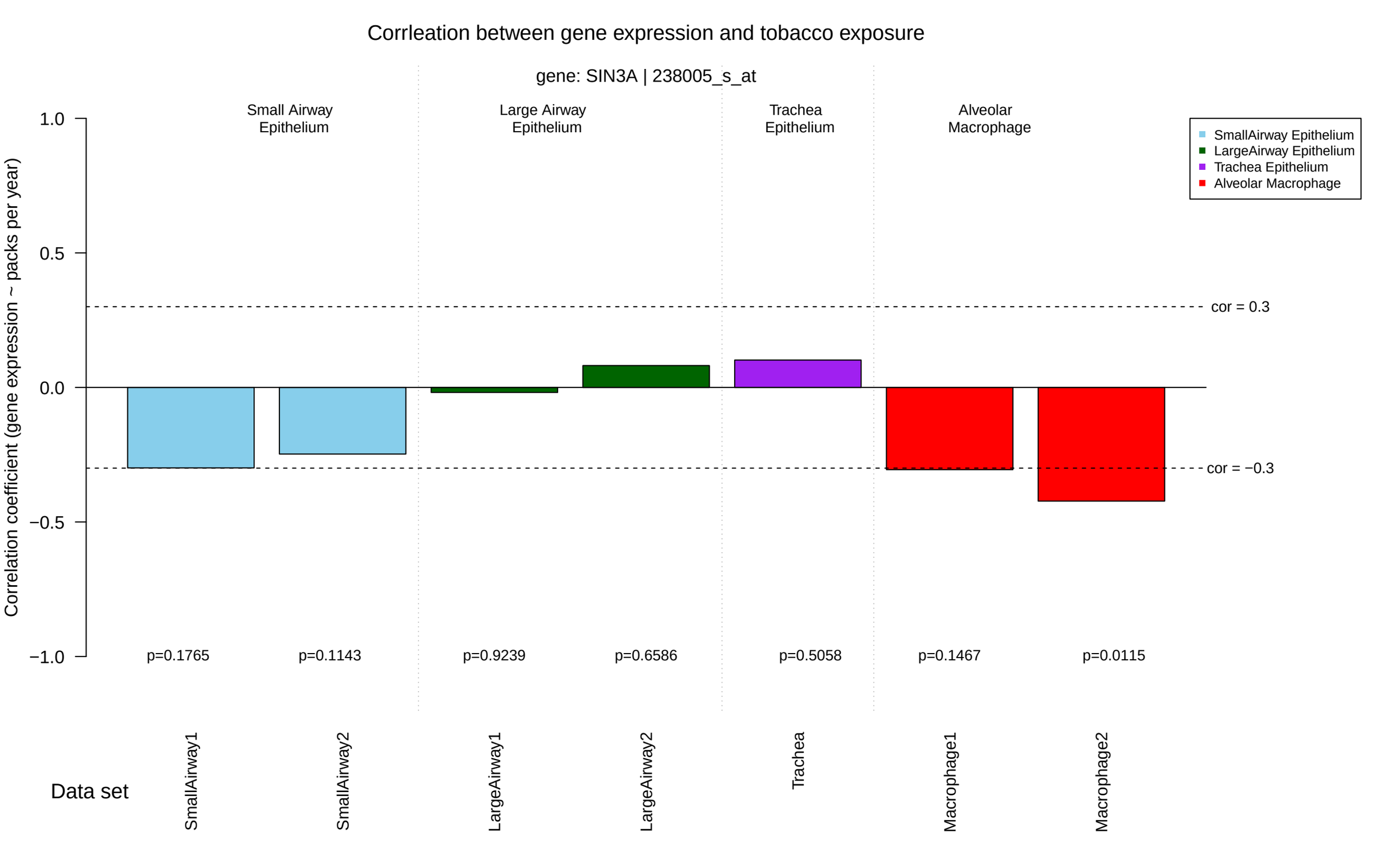

set: LargeAirway1 (p=0.7098)
SIN3A | 238005_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5394)
SIN3A | 238005_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2044)
SIN3A | 238005_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0765)
SIN3A | 238005_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0085)
SIN3A | 238005_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SIN3A | 238005_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1765
p=0.1143
p=0.9239
p=0.6586
p=0.5058
p=0.1467
p=0.0115
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2